

LOUVAIN
School of Management

LLSM2904 - Managing Complexity

Group project

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1 Part 1

1.1 Weekly Returns

For each asset with weekly price P_t observed at date t, we compute the weekly return using the log-return formula 1 instead of the classic return 2.

$$r_i = \ln(\frac{P_i}{P_{i-1}}) = \ln(P_i) - \ln(P_{i-1}) \tag{1}$$

$$r_i = \frac{P_i - P_{i-1}}{P_{i-1}} \tag{2}$$

Due to the formula, the first observation has no return because P_{-1} is unavailable. We use log-returns mainly because they behave more cleanly for this assignment:

1. **Time additivity**: multi-period log-returns add up exactly (weekly logs sum to a monthly log return over the same span).
2. **Consistency**: Consistency for losses and tail work using log returns keeps the loss definition $L_t = -r_t$ simple and scale-consistent across assets.
3. **Close to simple returns at weekly frequency**: For small weekly changes, log and simple returns are very similar. We do not lose interpretability but gain nicer aggregation properties.

Furthermore, we noticed that in the Gold series, observations from 01-01-1971 to 03-01-1975 are given on a monthly basis, with gaps of 4–5 weeks between consecutive quotes, affecting the mean, standard deviation and the scatter plots. To keep a consistent weekly-return definition, we converted each n-week return Rt into an equivalent constant weekly return r_t using geometric scaling:

$$(1 + r_t)^n = 1 + R_t$$
$$r_t = (1 + R_t)^{1/n} - 1,$$

with $n \in \{4, 5\}$ depending on whether the time gap is 4 or 5 weeks.

Given weekly returns $r_1, ..., r_T$, we computed the mean and the standard deviation using respectively the formulas 3 and 4 and obtained the digits in the table 1.

$$\overline{r} = \frac{1}{T}\sum_{t=1}^{T} r_t \tag{3}$$

$$\sigma = \sqrt{(\frac{1}{T-1}) \cdot \sum_{t=1}^{T}(r_t - \overline{r})^2} \tag{4}$$

	S&P500	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
Mean (in %)	0.149	0.190	0.167	0.134	1.726
Standard deviation (in %)	2.287	2.772	2.438	4.326	13.268

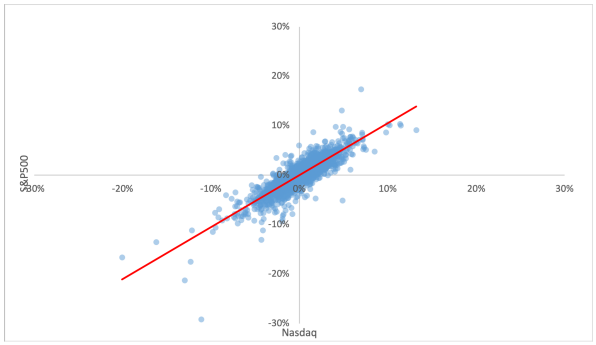
Table 1: Mean and Standard deviation of all the assets on a weekly basis

We can therefore say that Bitcoin is by far the highest-performing asset, but precaution is mandatory regarding its standard deviation. Then comes the Nasdaq index due to its link with the high-performing US tech sector. The least-performing asset is silver. If we take into account its high standard deviation, it does not look like a good investment.

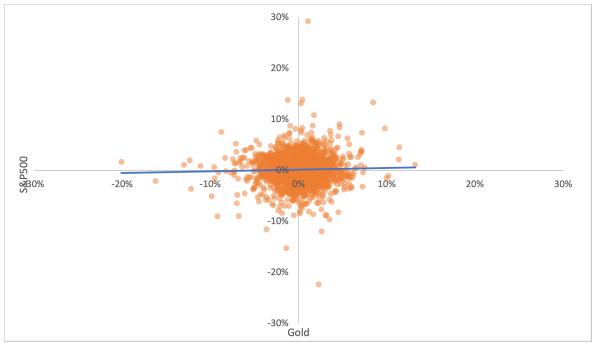
1.2 Scatter plots of returns

We created four scatter plots by using the dedicated to in Excel. The plots are represented in figure 1. On each one of them, a trendline is also drawn using a linear regression.

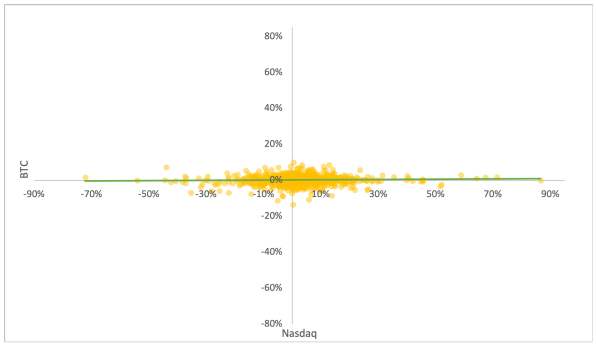
Several conclusions can be made. First, it is clear in Figure 1b and 1c that the S&P500 and gold, and Bitcoin and Nasdaq are respectively uncorrelated. Indeed, the trendlines on both graphs are almost horizontal, which means that there is no clear relation between those values. However, the S&P500 and Nasdaq, and gold and silver are very correlated. The slopes of the trendlines are respectively 1.051 and 1.251 for plots 1b and 1d. The R^2 are equal to 0.75 and 0.49, respectively as well. Therefore, both stock indexes are much more correlated than the two metals. This means that, statistically speaking, if one of the prices goes up or down, the other must follow in a similar proportion.



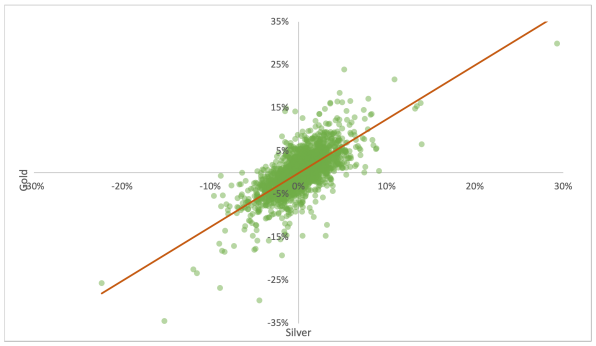
(a) S&P500 and Nasdaq



(b) S&P500 and Gold



(c) Bitcoin and Nasdaq



(d) Gold and Silver

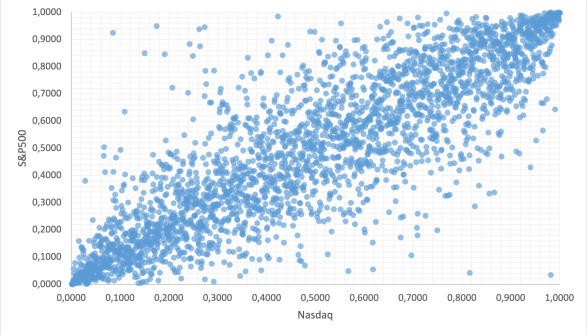
Figure 1: Scatter plots on weekly basis

1.3 Empirical copula plots

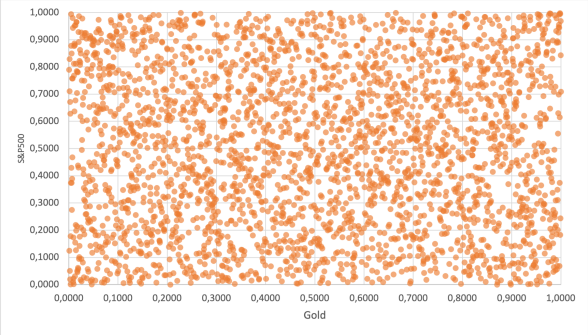
The figure 2 illustrates the empirical copula plots for the same couple of assets as in subsection 1.2.

Figures 2b and 2c confirm the conclusions of the previous subsection. Indeed, both those pairs of assets are almost perfectly uncorrelated, since their copula plots are uniform point clouds. In this case, the copula plots do not bring additional information.

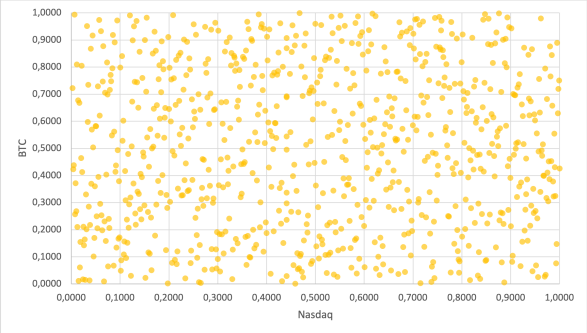
Nevertheless, the plots 2a and 2d give more details on the way the pairs of assets are linked. Indeed, it can be seen on both plots that the concentration of points is more important in the corners $(0, 0)$ and $(1, 1)$. This suggests that the asset values will react together when they increase or decrease rapidly, due to external shocks for example. This phenomenon is called "tail dependence" (Münnix & Schäfer, 2011). When the markets are stable, both values have room to evolve more independently. This is mainly the case for the metals (fig. 2d) where a lot of outliers can be seen (i.e. point far from the central ellipse).



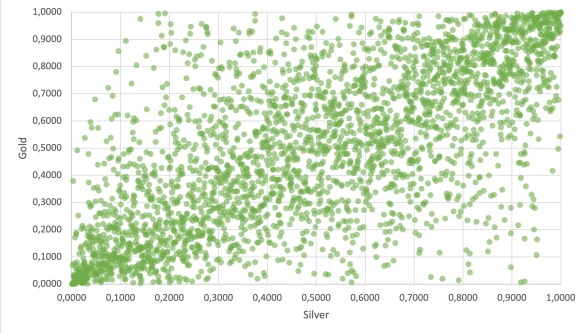
(a) S&P500 and Nasdaq



(b) S&P500 and Gold



(c) Bitcoin and Nasdaq



(d) Gold and Silver

Figure 2: Copula plots on a weekly basis

1.4 Risk measures

1.4.1 Value at risk

The values at risk (VaR) we computed are shown in table 2. We used two different ways to compute the VaR. The equation 5 to compute the *parametric* one.

$$\mathrm{VaR}_\alpha = \mu - z_\alpha \cdot \sigma, \tag{5}$$

with μ being the mean, σ being the standard deviation and z_α the quantile of the standard normal distribution corresponding to a specific confidence level.

The use of this formula implies the assumption that the data is normally distributed. This assumption is acceptable because of the Central Limit Theorem (CLT). The CLT states that the sum (or the mean) of a large number of independent and identically distributed random variables converges to a normal distribution, regardless of their original distribution. We can say that one discrete return is a random variable.

In order to compute the *empiric* VaR, since the VaR can be interpreted as "the smallest loss among the x% biggest", we simply use the excel function `-CENTILE.INCLURE()`. This function returns the value allocated at the desired (i.e. 5 or 1%) percentile.

	type of VaR (in %)	**S&P500**	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
VaR_{95}	Empirical	3.468	4.322	3.626	6.281	16.919
	Parametric	3.603	4.356	3.832	6.960	20.033
VaR_{99}	Empirical	6.608	7.864	6.433	12.078	37.215
	Parametric	5.181	6.269	5.515	9.945	29.188

Table 2: Comparison of Empirical and Parametric VaR (Weekly)

We can make several interpretations from those digits. First, both empirical and parametric VaR remain close. This confirms that the normal distribution assumption is acceptable. However, since we have less data about Bitcoin, the differences are a bit higher. Secondly, we can notice that the empirical computation always tends to minimize the risk. This illustrates that the assumption previously mentioned is a real approximation. In reality, the tails are fatter than a normal distribution because krachs are more frequent. Thirdly, we can also notice that proportionally the Var_{99} is greater than Var_{95} in the Bitcoin case than for the other assets (i.e., more than double). This illustrates that BTC is more subject to "catastrophic" events than the others. In other words, the worst events have higher losses for BTC than the worst events for the other assets. This is called exposure to Black Swans (Schneider, 2023).

We are able to create a first risk hierarchy between our assets:

$$\text{S\&P500} < \text{Gold} < \text{Nasdaq} < \text{Silver} < \text{Bitcoin},$$

This result was expected because we know that the S&P500 is less dependent on the technology sector. Both metals are a bit more risky since they are a single asset. Bitcoin is the most risky because it is a cryptocurrency.

However, using VaR to make this kind of ranking is not efficient because there is a big issue. VaR does not give any information about how the losses are distributed within those X% worst losses.

1.4.2 Expected shortfall

Expected shortfall (ES) solves this issue by taking the average of the X% of worst losses. As before, we compute the ES using two different methods. Concerning the "parametric" one, we used equation 6

$$ES_\alpha = \mu - \sigma \cdot \frac{\phi(z_\alpha)}{1-\alpha} \tag{6}$$

with μ being the mean, σ the standard deviation, α the confidence level, z_α the quantile of the standard normal distribution corresponding to α and $\phi()$ the pdf of the standard normal distribution. The use of this formula has the same implications as the VaR. Regarding the "empirical" ES, we simply compute the average of the losses using the function `-MOYENNE.SI(VaR;"<="all the data)`. The obtained values are shown in table 3.

	type of ES (in %)	**S&P500**	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
ES_{95}	Empirical	5.3931	6.4403	5.4615	10.1955	28.9515
	Parametric	4.6074	5.5738	4.9034	8.8600	25.8610
ES_{99}	Empirical	9.1070	11.2547	8.7742	18.5087	45.6613
	Parametric	5.8963	7.1356	6.2773	11.2973	33.3364

Table 3: Comparison of Empirical and Parametric ES (Weekly)

The main conclusion we can draw is that parametric computation minimizes risk, as mentioned before. This is especially the case for Bitcoin, where the difference reaches 12.3 points.

If we establish a second risk ranking based on the ES, it will be:

$$\text{Gold} \approx \text{S\&P500} < \text{Nasdaq} < \text{Silver} < \text{Bitcoin},$$

At this stage, gold is in first place with the S&P500. Indeed, they have comparable risk measures here and are both known as refuge assets in times of crisis. The rest of the assets remain in the same order.

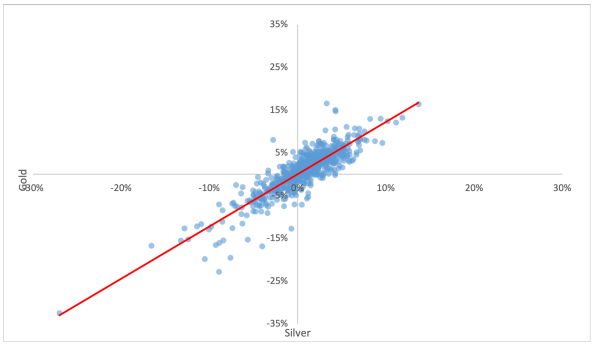
1.5 Monthly returns

To produce the data on a monthly basis, we used the pivot table tool provided by Excel. It allowed us to select the highest price for each asset for each month. Afterward, we made the same computations as before (i.e. log-return, basic statistics, plots, and risk measures). We decided to take the maximum price instead of the mean of each month because it is the simplest way to proceed. The difference between both methods is negligible.

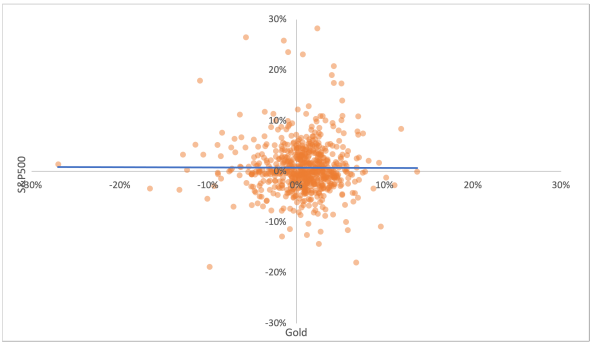
Table 4 shows the means and the standard deviations. Figures 3 and 4 refer to the scatter plots and copula plots respectively. The figures for VaR and ES are shown in table 5 and table 6 respectively.

	S&P500	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
Mean (%)	0.645	0.821	0.738	0.579	7.517
Standard deviation (%)	3.678	5.143	5.382	9.062	30.645

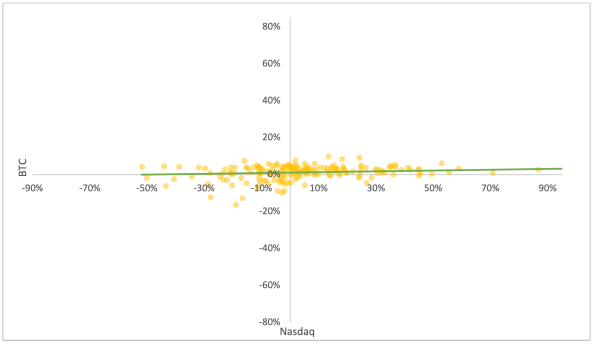
Table 4: Mean and Standard deviation of all the assets on the Monthly basis



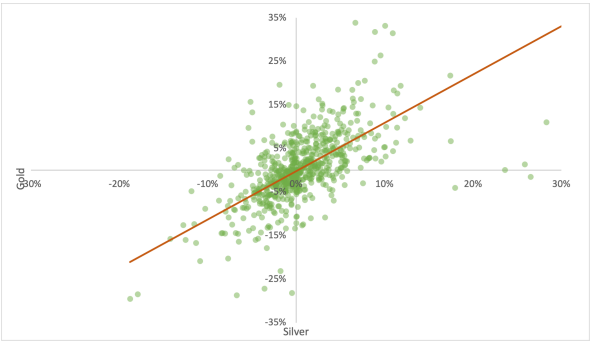
(a) S&P500 and Nasdaq



(b) S&P500 and Gold

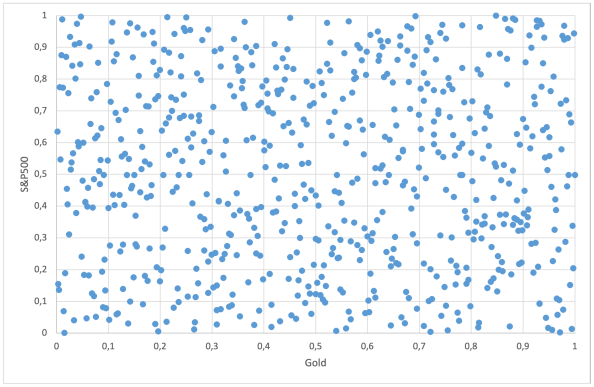


(c) Bitcoin and Nasdaq

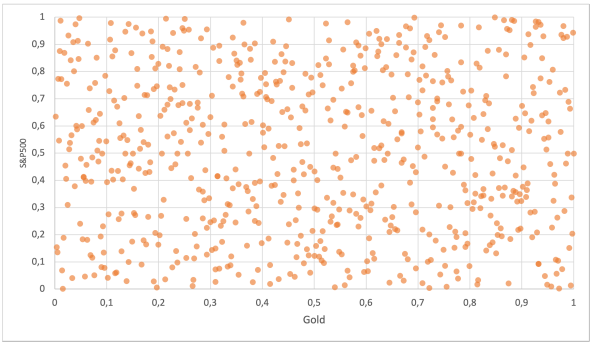


(d) Gold and Silver

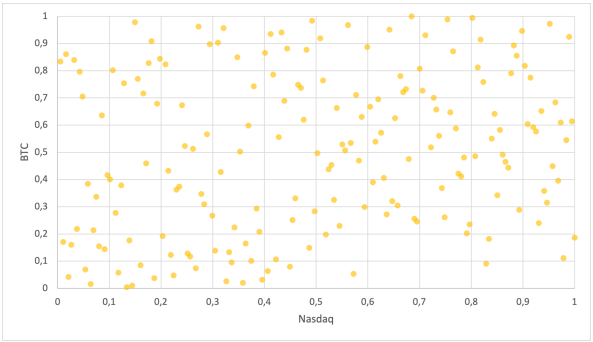
Figure 3: Scatter plots on a monthly basis



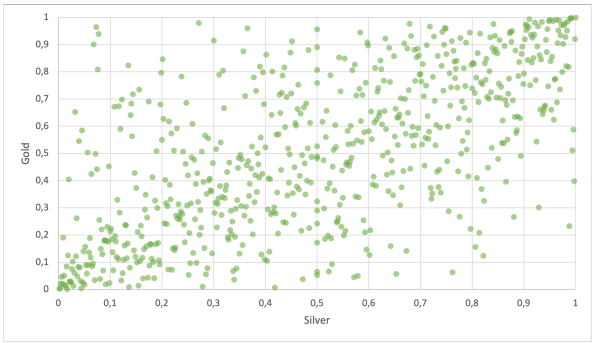
(a) S&P500 and Nasdaq



(b) S&P500 and Gold



(c) Bitcoin and Nasdaq



(d) Gold and Silver

Figure 4: Copula plots on a monthly basis

	type of VaR (in %)	**S&P500**	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
VaR_{95}	Empirical	5.422	7.437	6.235	12.362	28.621
	Parametric	5,387	7.614	8.088	14.282	42.741
VaR_{99}	Empirical	10.726	16.263	11.414	21.772	45.012
	Parametric	7.924	11.163	11.802	20.535	63.886

Table 5: Comparison of Empirical and Parametric VaR (Monthly)

	type of ES (in %)	**S&P500**	**Nasdaq**	**Gold**	**Silver**	**Bitcoin**
ES_{95}	Empirical	9.156	13.088	9.170	18.717	39.483
	Parametric	7.002	9.873	10.452	18.262	56.201
ES_{99}	Empirical	14.938	20.671	14.226	34.056	51.089
	Parametric	9.074	12.771	13.484	23.368	73.467

Table 6: Comparison of Empirical and Parametric ES (Monthly)

Switching the time frame has a different impact on our data. In the same way that changing the sampling frequency affects the noise on the remaining signal, the noise included in the monthly sample must be reduced compared to the weekly sample, from Mouraux and Mulders (2023). The interpretation could be the fact that here the system can lump a catastrophic week and the three other weeks of the month together. In other words, a dramatic return for a specific week can be erased by positive returns for the rest of the month. This has the effect of reducing the width of the tails and therefore enhancing the normality of the distribution. Overall, the approximation of the CLT should be more accurate (i.e. each data point is an independent random variable).

When we compare the basic statistics, two things appear. First, the mean of the monthly data is roughly 4 times that of the weekly data. Second, the monthly standard deviation is roughly the weekly one multiplied by a factor of $\sqrt{4}$. This is because the variance is additive with time and is the square of the standard deviation.

If we focus on the VaR, we can see that the monthly VaR is most of the time smaller than $\sqrt{4}$ times the weekly VaR. This confirms that a part of the "noise" is removed. Worst events are obscured. Proportionally, therefore, the tails are effectively thinner.

Regarding the ES, the transition to a monthly horizon further illustrates the smoothing of extreme tail events. Since ES measures the average loss beyond the VaR threshold, the observed values are generally lower than the weekly ES scaled by 4 This confirms that a "catastrophic week" is often offset by the other three weeks of the month. Consequently, the monthly ES reflects a distribution with thinner tails, where the impact of outliers is dampened by time aggregation, aligning more closely with the Central Limit Theorem's predictions.

If we interpret the eight plots now, several facts appear too. On a monthly basis, the correlation between the S&P 500 and Nasdaq is even higher ($R^2 = 0,765$ and $0,752$ on a weekly basis). This illustrates that, besides the crisis periods (i.e. periods where all assets fluctuate in the same direction), both assets remain aligned. This is proof of a more structural link between them (i.e. the fact that 84 companies are in both indexes). This is also visible in figure 4c since there is no more concentration of points in the corners. On the other hand, the metals are less correlated ($R^2 = 0.434$ only) and their points remain concentrated on the copula (fig. 4d).

2 Part 2

2.1 Tail index estimation

We estimate the tail index using the linear regression approach presented in the lecture about the chapter 4. The results of our computations are shown in figure 5.

Let $L_t = -r_t$ denote weekly losses. We focus on the upper tail of the loss distribution, corresponding to large positive losses. Assuming a Pareto-type tail behavior:

$$1 - F(x) \approx kx^{-\alpha}, \tag{7}$$

, taking logarithms leads to:

$$-\log(1 - F(x)) = -\log(k) + \alpha \log(x). \tag{8}$$

This implies that a linear relationship should hold between $-\log(1 - F(x))$ and $\log(x)$, where the slope provides an estimate of the tail index α. In practice, we approximate $1 - F(x)$ by the empirical proportion of observations exceeding x, i.e.:

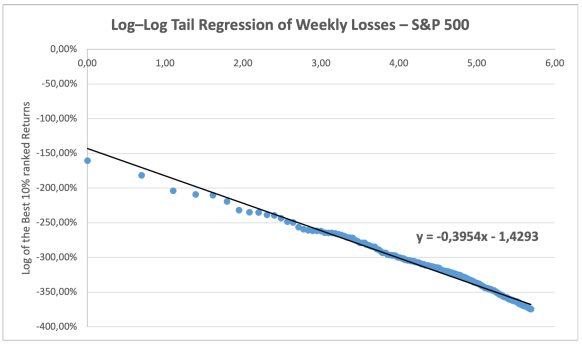
$$1 - F(x_i) \approx \frac{i}{n}, \tag{9}$$

where i is the rank of the observation in the ordered sample of losses (from largest to smallest), and n is the total number of observations.

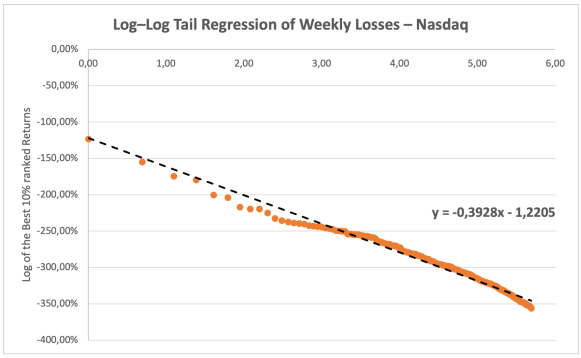
This choice reflects a trade-off between bias and variance: selecting too few observations leads to unstable estimates, while selecting too many observations introduces non-tail data and biases the estimation. The slope of the regression line provides an estimate of the tail index α, while the intercept corresponds to $-\log(k)$.

Interpretation of the regression coefficients

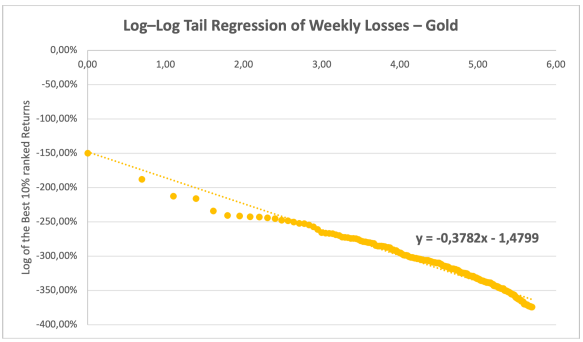
The log-log regression provides an estimate of the tail behavior of the loss distribution. The slope of the regression line corresponds to the tail index α (up to sign). In our case, the slope is negative due to the specification used, and the tail index is given by its absolute value. A lower value of α indicates heavier tails, meaning that extreme losses are more likely to occur. Conversely, a higher value of α corresponds to thinner tails and lower tail risk. The intercept captures the scale of the tail distribution but does not directly affect the heaviness of the tail. It mainly shifts the regression line vertically and reflects the overall magnitude of tail probabilities.



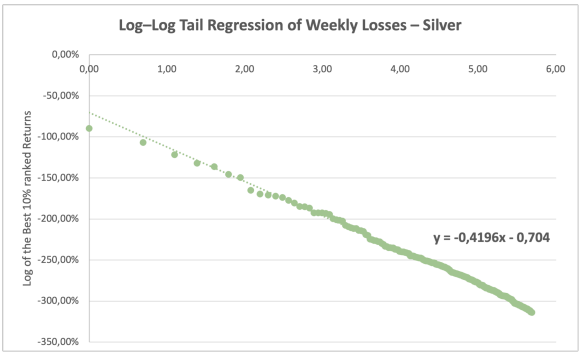
(a) S&P500



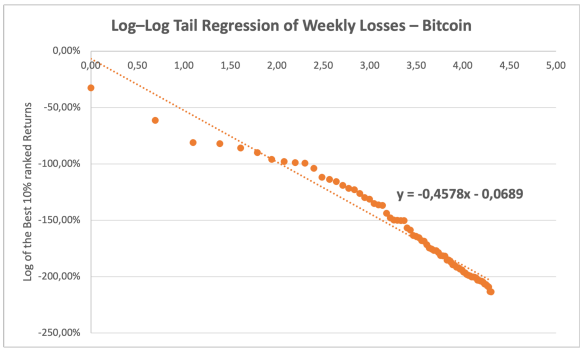
(b) Nasdaq



(c) Gold



(d) Silver



(e) Bitcoin

Figure 5: Log-log tail regression for weekly losses across assets

Comparison across assets

The estimated slopes differ across assets, indicating different levels of tail risk. Bitcoin exhibits the steepest slope in absolute value, corresponding to the lowest tail index. This indicates very heavy tails and a high probability of extreme losses. Silver and Nasdaq also display relatively heavy tails, with intermediate tail index values. Gold shows a slightly higher tail index, suggesting thinner tails and lower exposure to extreme events. The S&P500 lies between these assets, with moderate tail risk.

To conclude, Bitcoin appears to be the riskiest asset in the tail, as it has the lowest estimated tail index and therefore the highest likelihood of extreme losses. Traditional assets such as Gold and the S&P500 exhibit less extreme tail behavior, making them comparatively safer in terms of tail risk. These findings are consistent with the higher volatility typically observed in cryptocurrency markets.

2.2 GPD estimation using the Expected Value property

To model the tail behavior of weekly losses, we use the mean excess function approach. The mean excess function is defined as:

$$e(u) = E[X - u \mid X > u]. \tag{10}$$

For a Generalized Pareto Distribution (GPD), this function is linear in u:

$$e(u) = \frac{\beta + \xi u}{1 - \xi}. \tag{11}$$

Thus, plotting $e(u)$ against u allows us to assess whether the GPD model is appropriate and to estimate the parameters.

Estimation procedure

For each asset, several high thresholds u were selected in the upper tail of the weekly loss distribution. For each threshold, exceedances were computed as:

$$y_i = X_i - u \quad \text{for all } X_i > u, \tag{12}$$

where X_i denotes the weekly losses. The mean excess function was then estimated as:

$$e(u) = E[X - u \mid X > u]. \tag{13}$$

In practice, this is computed as the sample average of exceedances:

$$e(u) = \text{average}(X_i - u). \tag{14}$$

The mean excess values were plotted against the corresponding thresholds u to produce the mean excess plot.

Parameter estimation

Under the GPD assumption, the mean excess function is linear in u:

$$e(u) = a + bu \tag{15}$$

Using linear regression, the parameters a and b are estimated from the mean excess plot. The GPD parameters are then obtained as follows:

$$\xi = \frac{b}{1+b} \tag{16}$$

$$\beta = a(1-\xi), \tag{17}$$

where ξ is the shape parameter and β is the scale parameter of the GPD distribution. The slopes obtained are in the table 7. Positive slopes (Nasdaq, Silver, Gold) indicate $\xi > 0$, confirming heavy tails. Negative slopes (Bitcoin, S&P500) suggest that the GPD assumption may not hold perfectly over the selected threshold range, or that the estimation is affected by sampling variability.

Asset	**Slope**
Nasdaq	0.2506
Silver	0.3352
Gold	0.0917
Bitcoin	-0.2769
S&P500	-0.1051

Table 7: Slope coefficients for GDP by expected value property

The shape parameter ξ determines the heaviness of the tail of the loss distribution. A positive value of ξ ($\xi > 0$) indicates a heavy-tailed distribution, meaning that extreme losses occur with non-negligible probability. Moreover, larger values of ξ imply a heavier tail, and therefore more severe and more frequent extreme losses. Conversely, values of ξ close to zero correspond to thinner tails, closer to the normal distribution, while negative values would indicate a bounded tail.

The results obtained from this method were compared with the tail index estimates derived from the linear regression approach in Question 1. This comparison provides a consistency check between the two Extreme Value Theory (EVT) techniques. Overall, the assets with higher estimated ξ are those exhibiting the greatest exposure to extreme losses, confirming the conclusions obtained previously using the tail index.

The results are partially consistent with the tail index estimation. Assets with heavy tails in Question 1 (such as Nasdaq and Silver) also exhibit increasing mean excess functions, supporting the presence of extreme risk. However, the inconsistency observed for Bitcoin and S&P500 highlights the sensitivity of the mean excess method to the choice of threshold and the limited number of extreme observations.

To sum up, the mean excess plots suggest that most assets exhibit heavy-tailed behavior, with particularly strong evidence for Silver and Nasdaq. Bitcoin remains a highly

risky asset according to Question 1, but its mean excess plot indicates that further refinement of the threshold selection may be needed. These results confirm that extreme value methods are sensitive to data selection and should be interpreted with caution.

2.3 GPD estimation using Maximum Likelihood and tail risk measures

Data preparation

The analysis is conducted on weekly returns computed in Part 1. Losses are defined as:

$$L_t = -r_t \tag{18}$$

so that large positive values correspond to large losses. All tail analyses are performed on the upper tail of this loss distribution.

Threshold selection

To fit the GPD model, a high threshold u was selected for each asset, corresponding approximately to the 90% percentile of weekly losses. This choice ensures that only extreme losses are modeled and a sufficient number of exceedances remain for reliable estimation.

Let n_u be the number of exceedances and N the total number of observations. The exceedance probability is defined as:

$$p_u = \frac{n_u}{N} \tag{19}$$

In our dataset, this corresponds to approximately 287 exceedances out of 2869 observations, which provides a good balance between bias and variance.

Maximum Likelihood estimation

The exceedances are defined as:

$$y_i = L_i - u \quad \text{for } L_i > u, \tag{20}$$

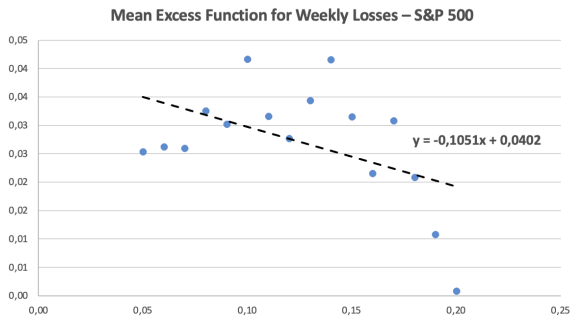
and the GPD density function as :

$$f(y) = \frac{1}{\beta}\left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi - 1}, \tag{21}$$

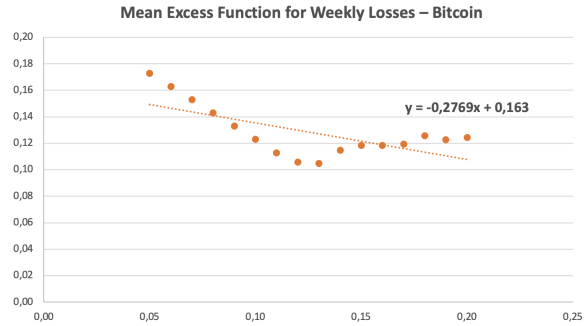
where ξ is the shape parameter and β is the scale parameter. The parameters are estimated by maximizing the log-likelihood:

$$L(\xi, \beta) = -n_u \ln(\beta) - \left(1 + \frac{1}{\xi}\right) \sum \ln\left(1 + \frac{\xi y_i}{\beta}\right) \tag{22}$$

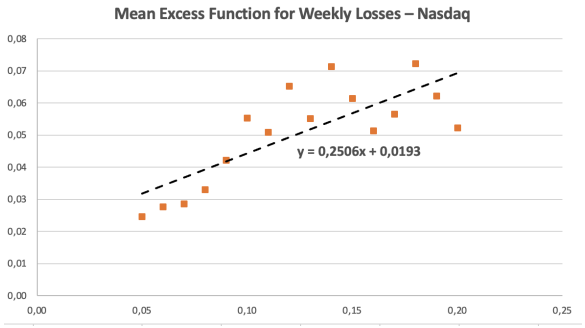
The maximization was performed using the Excel Solver nonlinear optimization tool. At the end, the estimated parameters are those in the table 8.



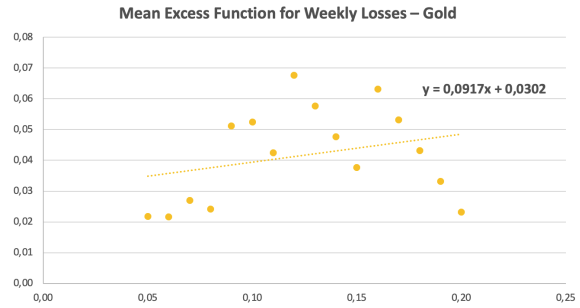
(a) S&P500



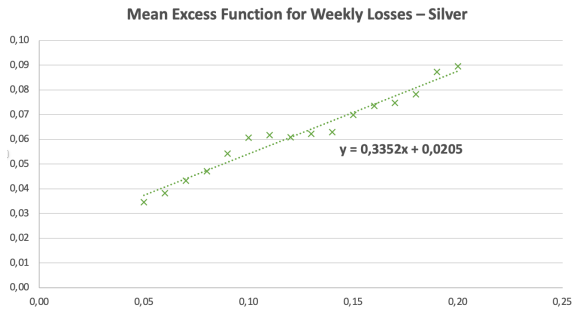
(b) Nasdaq



(c) Gold



(d) Silver



(e) Bitcoin

Figure 6: Mean excess functions for weekly losses across assets

Asset	ξ **(Shape)**	β **(Scale)**	**Log Likelihood**
S&P500	0.19	0.01	885.03
Nasdaq	0.94	0.0001	6468.83
Gold	0.08	0.02	863.70
Silver	0.80	0.001	1899.77
Bitcoin	0.17	0.08	105.77

Table 8: Estimated GPD parameters (ξ and β)

The shape parameter ξ determines the heaviness of the tail. $\xi > 0$ indicates a heavy-tailed distribution and larger values imply more severe extreme losses. Nasdaq and Silver exhibit the highest ξ, indicating extremely heavy tails and high exposure to extreme losses. Bitcoin and S&P500 show moderate tail risk, while Gold exhibits the lowest ξ, suggesting thinner tails. These results are broadly consistent with the tail index estimation and mean excess method, although some differences arise due to estimation sensitivity.

Using the fitted GPD model, we compute the unconditional tail risk measures. These measures were computed for 95% and 99% confidence level with as formula :

$$VaR_q = u + \frac{\beta}{\xi}\left[\left(\frac{1-q}{p_u}\right)^{-\xi} - 1\right] \tag{23}$$

$$ES_q = \frac{VaR_q}{1-\xi} + \frac{\beta - \xi u}{1-\xi} \tag{24}$$

The GPD-based VaR and ES were compared to empirical estimates obtained in section 1. We observe that, first GPD-based VaR and ES are generally higher than empirical values. Second, this reflects the heavy-tailed nature of financial losses. Third, empirical methods underestimate extreme risk due to limited tail observations. The differences are particularly strong for Nasdaq, Silver and Bitcoin, which exhibit the heaviest tails.

The results depend on the choice of threshold u and the number of exceedances. If u is too low, the GPD assumption may not hold; if it is too high, the estimation becomes unstable due to fewer observations. This highlights the importance of robustness checks in extreme value analysis.

To conclude, the MLE-based GPD estimation confirms the presence of heavy tails in weekly losses. Nasdaq and Silver appear to be the riskiest assets in the tail, followed by Bitcoin. Gold and S&P500 show comparatively lower tail risk. Overall, Extreme Value Theory provides a more realistic framework for modeling extreme losses than empirical methods, particularly for rare and severe events.

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